

Zhuowei Chen

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EDUCATION

University of Pittsburgh Ph.D. in Computer Science Advisor: Prof. Xiang Lorraine Li	Pittsburgh, PA Sep 2025 - Current
Guangdong University of Foreign Studies B.E. in Software Engineering Advisor: Prof. Lianxi Wang	Guangzhou, China Sep 2021 - Jun 2025
University of California, Berkeley Berkeley Visiting Student	Berkeley, CA Aug 2023 - Jan 2024

PUBLICATIONS

* represents equal contributions and † represents the corresponding author.

- Zhuowei Chen**, Bowei Zhang, Nankai Lin, Tian Hou, Lianxi Wang.
[Unlocking LLM Safeguards for Low-Resource Languages via Reasoning and Alignment with Minimal Training Data.](#)
The Fifth Workshop on Multilingual Representation Learning, MRL Workshop @ EMNLP 2025.
Reinforcement Learning LLM Safety
- Zhuowei Chen**, Qiannan Zhang, Shichao Pei.
[Injecting Universal Jailbreak Backdoors to LLMs in Minutes.](#)
The Thirteenth International Conference on Learning Representations, ICLR 2025.
Model Editing LLM Safety
- Zhuowei Chen**, Yuben Wu, Xinfeng Liao, Yujia Tian, Lianxi Wang[†].
[An Effective Deployment of Diffusion LM for Data Augmentation in Low-Resource Sentiment Classification.](#)
The 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024.
Generative Data Augmentation Diffusion LM
- Lianxi Wang, Yujia Tian*, **Zhuowei Chen***[†].
[Enhancing Hindi Feature Representation Through Fusion of Dual-Script Word Embeddings.](#)
The Joint Conference on Computational Linguistics, Language Resources and Evaluation, LREC-COLING 2024.
MoE Representation Enhancement Low-Resource Languages
- Zhuowei Chen**, Yujia Tian, Lianxi Wang[†], Shengyi Jiang.
[A Distantly-Supervised Relation Extraction Method Based on Selective Gate and Noise Correction.](#)
The 22nd China National Conference on Computational Linguistics, CCL 2023.
Feature Selection Noise Reduction
- Zhuowei Chen**, Liwei Chen, Christian Schunn, Raquel Coelho, Xiang Lorraine Li.
Neuron-Aware Active Few-Shot Learning for LLMs.
Under review @ ARR.
- Xinfeng Liao, Xuanqi Chen, Lianxi Wang, Jiahuan Yang, **Zhuowei Chen**, Ziyang Rong.
OTESGN: Optimal Transport Enhanced Syntactic-Semantic Graph Networks for Aspect-Based Sentiment Analysis.
The 25th IEEE International Conference on Data Mining, ICDM 2025
- Lianxi Wang, Yujia Tian*, **Zhuowei Chen***, Mutong Li, Nankai Lin[†].
EditMDS: An Iterative Optimization Method for Multi-Document Summarization Based on Edit Operations.
Data Intelligence.

EXPERIENCE

University of Pittsburgh Graduate Student Researcher Supervisor: Prof. Xiang Lorraine Li	Pittsburgh, PA Sep 2025 – Current
- Automatic Annotation Tool for Educational Peer-Feedback - Benchmarked six LLMs on annotation task across paradigms, including Zero-Shot, Few-Shot, Similarity RAG, Prompt Tuning, LoRA, Instruction Tuning, GRPO-based RL, and SFT-GRPO fused RL. - Developed a neuron-aware data selection framework for SFT that matched full-dataset accuracy with an 11% subset, demonstrating the utility of internal learning dynamics for data pruning.	

University of Massachusetts Boston*Research Intern*

Supervisor: Prof. Shichao Pei

Boston, MA

Mar 2024 – Oct 2024

- JailbreakLLM: Exploring Novel Jailbreak Backdoor Attacks on LLMs.
 - Proposed a novel method to inject universal backdoors into LLMs without additional datasets or extensive computational overhead (lowest 5 samples with 30 seconds editing).
 - Executed comprehensive experiments, confirming a high jailbreak success rate (over 90% on Llama2-7b) and highlighting the urgency for advanced defensive strategies in LLMs.

AI Lab, Wisers Information Ltd.*NLP Research Intern*

Hong Kong, China

Dec 2023 – Mar 2024

- Hong Kong Tourism Index Formulation
 - Built Roberta-based textual classification models with human-annotated social media content.
 - Applied transformers for time series regression to predict regional arrivals.

Guangzhou Key Laboratory of Multilingual Intelligent Processing*Undergraduate Research Student*

Supervisor: Prof. Lianxi Wang

Guangzhou, China

Nov 2021 – Mar 2024

- Deploying Diffusion LM for Data Augmentation in Text Classification.
 - Fine-tuned LMs with a diffusion objective to capture in-domain knowledge and generate samples by reconstructing label-related tokens.
 - Designed attention-based mask schedule for the diffusion LM, balancing domain consistency, label consistency, and context diversity.
 - Conducted analyses and visualizations to study its underlying mechanism, followed by experiments validating its effectiveness across various low-resource scenarios.
- Enhancing Hindi Representations via Fusion of Pre-trained Language Models.
 - Proposed a method to enhance Hindi feature representation by combining Devanagari and Romanized Hindi pre-trained language models.
 - Ablations and extensive NLU task experiments show the superiority of our method, demonstrating the potential of multi-script integration to enhance low-resource language models.
- Distantly Supervised Relation Extraction (DSRE) with Learning-with-Noise Methods.
 - Combined selective gate and noise correction training framework for DSRE, which performs data selection and corrects noise labels during a three-stage training process.
 - Experiments demonstrated state-of-the-art performance, revealing a promising new approach for applying training-with-noise techniques in NLP.
- Multi-Label Text Classification (MLTC) with Knowledge Augmentation and Span Prediction.
 - Integrated span-prediction with an GNN-based knowledge augmentation module to enhance MLTC.
 - Conducted visualizations and analyses to study its working mechanism, emphasizing the critical role of incorporating domain-specific knowledge for LM.

SELECTED HONORS

- **Top Ten Outstanding Youth Award** (Top 10/30000) Guangdong University of Foreign Studies, 2025
- **China National Scholarship** (Top 0.2%) Ministry of Education of the PRC, 2024

SERVICES

- **Reviewer.** ACL Rolling Review, ICLR 2026, Data Intelligence.